Q.1) Write a PHP program to create a class temperature which contains data members as
Celsius and Fahrenheit . Create and Initialize all values of temperature object by using
parameterized constructor . Convert Celsius to Fahrenheit and Convert Fahrenheit to
Celsius using member functions. Display conversion on next page.

→
```
<?php
session_start();

// Define Temperature class
class Temperature {
   public $celsius;
   public $fahrenheit;

   // Parameterized constructor
   public function __construct($celsius, $fahrenheit) {
      $this->celsius = $celsius;
      $this->fahrenheit = $fahrenheit;
   }

   // Convert Celsius to Fahrenheit
   public function convertCelsiusToFahrenheit() {
      return ($this->celsius * 9/5) + 32;
   }

   // Convert Fahrenheit to Celsius
   public function convertFahrenheitToCelsius() {
      return ($this->fahrenheit - 32) * 5/9;
   }
}

// Handle form submission
if ($_SERVER['REQUEST_METHOD'] == 'POST') {
   // Retrieve input
   $c = $_POST['celsius'];
   $f = $_POST['fahrenheit'];

   // Create object
   $temp = new Temperature($c, $f);

   // Store values and results in session
   $_SESSION['celsius'] = $c;
   $_SESSION['fahrenheit'] = $f;
   $_SESSION['converted_f'] = $temp->convertCelsiusToFahrenheit();
   $_SESSION['converted_c'] = $temp->convertFahrenheitToCelsius();

   // Redirect to result page
```

```
    header("Location: temperature_result.php");
    exit();
}
?>

<!DOCTYPE html>
<html>
<head>
    <title>Temperature Input</title>
</head>
<body>
    <h2>Enter Temperature Values</h2>
    <form method="post">
        <label>Celsius:</label>
        <input type="number" step="any" name="celsius" required><br><br>

        <label>Fahrenheit:</label>
        <input type="number" step="any" name="fahrenheit" required><br><br>

        <input type="submit" value="Convert">
    </form>
</body>
</html>

<?php
session_start();
?>

<!DOCTYPE html>
<html>
<head>
    <title>Temperature Conversion Result</title>
</head>
<body>
    <h2>Temperature Conversion Results</h2>

    <p><strong>Entered Celsius:</strong> <?php echo $_SESSION['celsius']; ?> °C</p>
    <p><strong>Converted to Fahrenheit:</strong> <?php echo round($_SESSION['converted_f'], 2);
?> °F</p>

    <hr>

    <p><strong>Entered Fahrenheit:</strong> <?php echo $_SESSION['fahrenheit']; ?> °F</p>
    <p><strong>Converted to Celsius:</strong> <?php echo round($_SESSION['converted_c'], 2); ?>
°C</p>
</body>
</html>
```

Q.2) Write a script to create XML file ‘University.xml’. The element details of ‘University.x
ml’ Are as

follows:
<Univ>
<Uname>----------</Uname>
<CITY>----------</CITY>
<Rank>----------<</Rank>
</Univ>

a) Store the details of at least 3 universities.

b) Link the 'University.xml' file to CSS and get well formatted output as given below.

i) Uname : Color:black;
Font-family: copperplate G0thic Light;
Font size: 16pt; Font:Bold; ii) City and Rank Color:Yellow;
Font-family: Arial;
Font-size : 12pt;
Font: Bold;

→ <?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/css" href="university_style.css"?>

```
<Universities>
   <Univ>
      <Uname>Oxford University</Uname>
      <CITY>Oxford</CITY>
      <Rank>1</Rank>
   </Univ>

   <Univ>
      <Uname>Harvard University</Uname>
      <CITY>Cambridge</CITY>
      <Rank>2</Rank>
   </Univ>

   <Univ>
      <Uname>Stanford University</Uname>
      <CITY>Stanford</CITY>
      <Rank>3</Rank>
   </Univ>
</Universities>
```

Css

```
Uname {
   color: black;
   font-family: "Copperplate Gothic Light", sans-serif;
   font-size: 16pt;
   font-weight: bold;
   display: block;
   margin-bottom: 4px;
}
```

```
CITY, Rank {
   color: yellow;
   font-family: Arial, sans-serif;
   font-size: 12pt;
   font-weight: bold;
   display: block;
   margin-bottom: 10px;
}

Univ {
   margin-bottom: 20px;
   display: block;
   background-color: #333;
   padding: 10px;
   border-radius: 5px;
}
```

Out put
Oxford University
Oxford
1

Harvard University
Cambridge
2

Stanford University
Stanford
3